Inductive power transfer

The inductive power transfer system used in micro-mobility elements is wireless, implying that the secondary coil present in the vehicle can be integrated with the battery pack in a completely sealed waterproof way. The whole solution is a controlled power supply that can talk directly to the battery management system on the vehicle itself.

Wireless inductive power transfer provides following benefits:

- Frictionless rotation with robustness of pulling out and coiling in of cable
- Hermetically sealed system to prevent corrosion and damage of parts
- Waterproof system that can be cleaned by high-pressure water
- Meet the present-day charging requirements and in addition has:
  - Phone allocation giving status of charge
  - Status of the batteries
  - Number of charges and discharges
  - Possibility to charge gently to improve battery life

Infrastructure required

Small electronic devices (having secondary coils) are attached to the vehicles, which are capable of transferring high amounts of power. This power transfer is done wirelessly as the software used configures the right DC voltage directly to the battery pack.

Wireless charging is said to be less efficient but if power loss that occurs between the power grid and a wired charging point of the batteries is considered, the efficiency is a bit better. The communication link present between the coils can regulate the frequencies of the coils' inductive field, and with this datalink interaction with the battery is easy.

Therefore, with the communication link, it is easier to deliver the correct voltage and current. Misalignments or problems can also be dealt with by adjustment in frequency to maintain high efficiency. One of the unique features in wireless charging is that the energy can be pushed back to the grid, allowing a bidirectional flow of power over the same set of coils. With such benefits available, wireless charging is definitely better suited for charging batteries than a conductive solution.

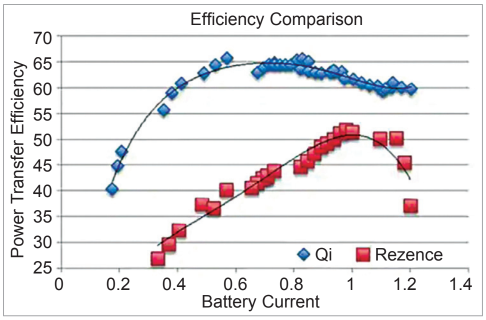

Fig. 2: The graph depicts the power transfer efficiency associated with closely-coupled inductive wireless charging system (Qi specification) and a resonant system (Rezence specification)

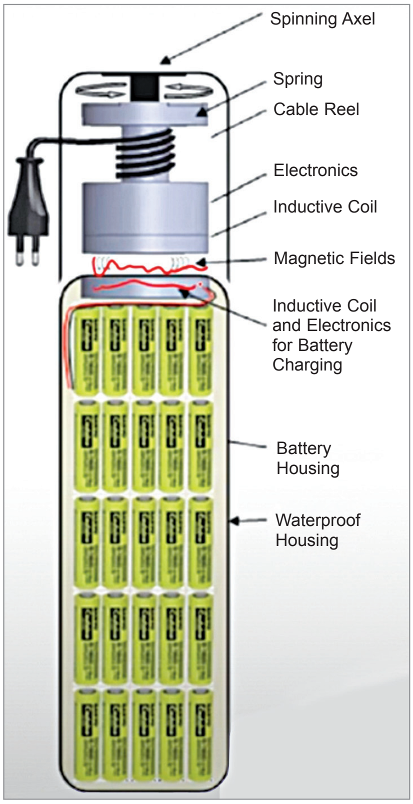

Fig. 3: Completely inductive power transfer system (Courtesy: MobiDock)

At the very core of the latest wireless technologies, a primary connector is made to send current to its coil to generate a magnetic field. This induces currents in the coil at the secondary connector. Both the coils are then precisely adjusted and trimmed for each product type and the magnetic fields are guided by a ferrite for maximum efficiency.

The strong magnetic field between both the connectors is used to transfer power. The primary and secondary connectors connect to each other using a very short range Wi-Fi that is designed to transfer data in the centre of this magnetic field. Ethernet user data is transmitted with low latency. The connectors also offer an RS232 or RS485 link using a separate antenna, and a dedicated internal communication link is present for power regulation and other system information.

The connectors can be controlled and be operated as a battery charger. Moreover, they have a two-way high-speed regulation channel that enables bidirectional power transfer. Hence, the same technology can be used to charge a battery as well as deliver energy from the battery as an inductive charger.

It would be interesting to know that 90% efficiency is achieved in most of the latest wireless power transfer systems, which is quite a fast rate of charging. The biggest advantage offered by the wireless systems is that they can control the batteries to prevent fire and can also report temperature fluctuations through the data link, so that the Li-ion cells can be disconnected if a risk of thermal runaway is found.

All these safety features, along with high efficiency, tick all boxes in terms of sustainability and proper use of energy. The automotive brands are now also focusing on developing compact and efficient secondary electronics for drones and other cost-sensitive products.

Infrastructure needed

For proper usage and charging of the vehicles an infrastructure that is maintained by a central server is needed. Micro-mobility startups and companies are developing solutions to manage shared mobility better.

Felyx, a Belgian micro-mobility startup, has developed a mobile app to locate, reserve, and connect with an e-scooter for urban transportation.

Duckt, another smart mobility startup, creates two-wheeler charging and parking stations that can be managed over an app.

Ground Control Systems, an American micro-mobility startup, provides 'charge and park' solutions over various campuses.